

MANAGEMENT OF BURNS IN THE CHILDREN'S EMERGENCY (THERMAL, ELECTRICAL & CHEMICAL)

MANAGEMENT OF MINOR – MODERATE THERMAL BURNS

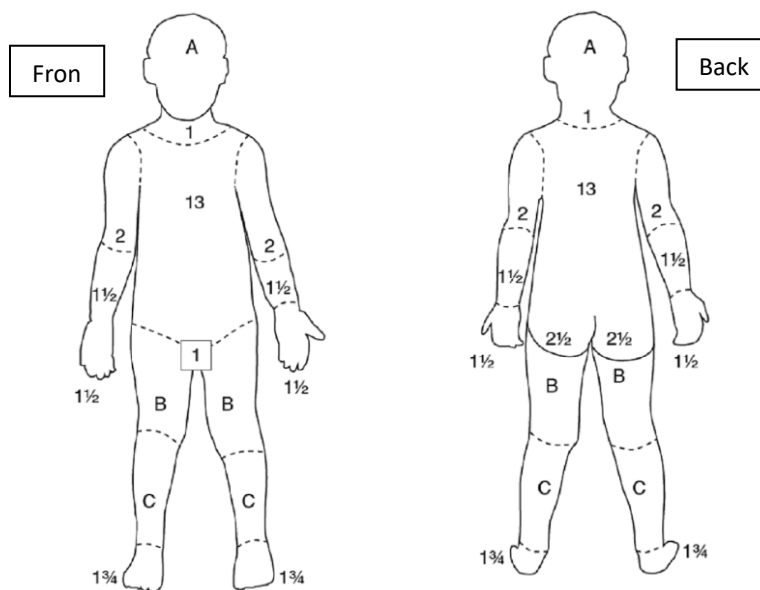
BURN DEPTH

	DEPTH	CHARACTERISTICS	HEALING TIME	PROGNOSIS	ANALGESIA
Superficial	Epidermis	Dry, erythematous	1 – 2 weeks	No scarring	PO analgesia
Partial thickness, superficial	Upper dermis	Wet, blisters Blanching	2 – 3 weeks	Non-scarring	IN fentanyl KIV IV morphine
Partial thickness, deep	Lower dermis	Dry Non-blanching	3 weeks – 2 months	Heals with contractures	IN fentanyl KIV IV morphine
Full thickness	Subcutaneous fat	White-yellow, leathery, painless	> 2 months	Contractures & scarring	IN fentanyl KIV IV morphine

EXTENT OF BURN (%TBSA)

- Superficial burns not included in %TBSA calculation:
 - Lund and Browder Chart – Highest inter-operator reliability
 - Palm method – Useful for small irregular burns
 - Rule of "9"s – Overestimates %TBSA in patients < 15-years-old

LUND AND BROWDER CHART



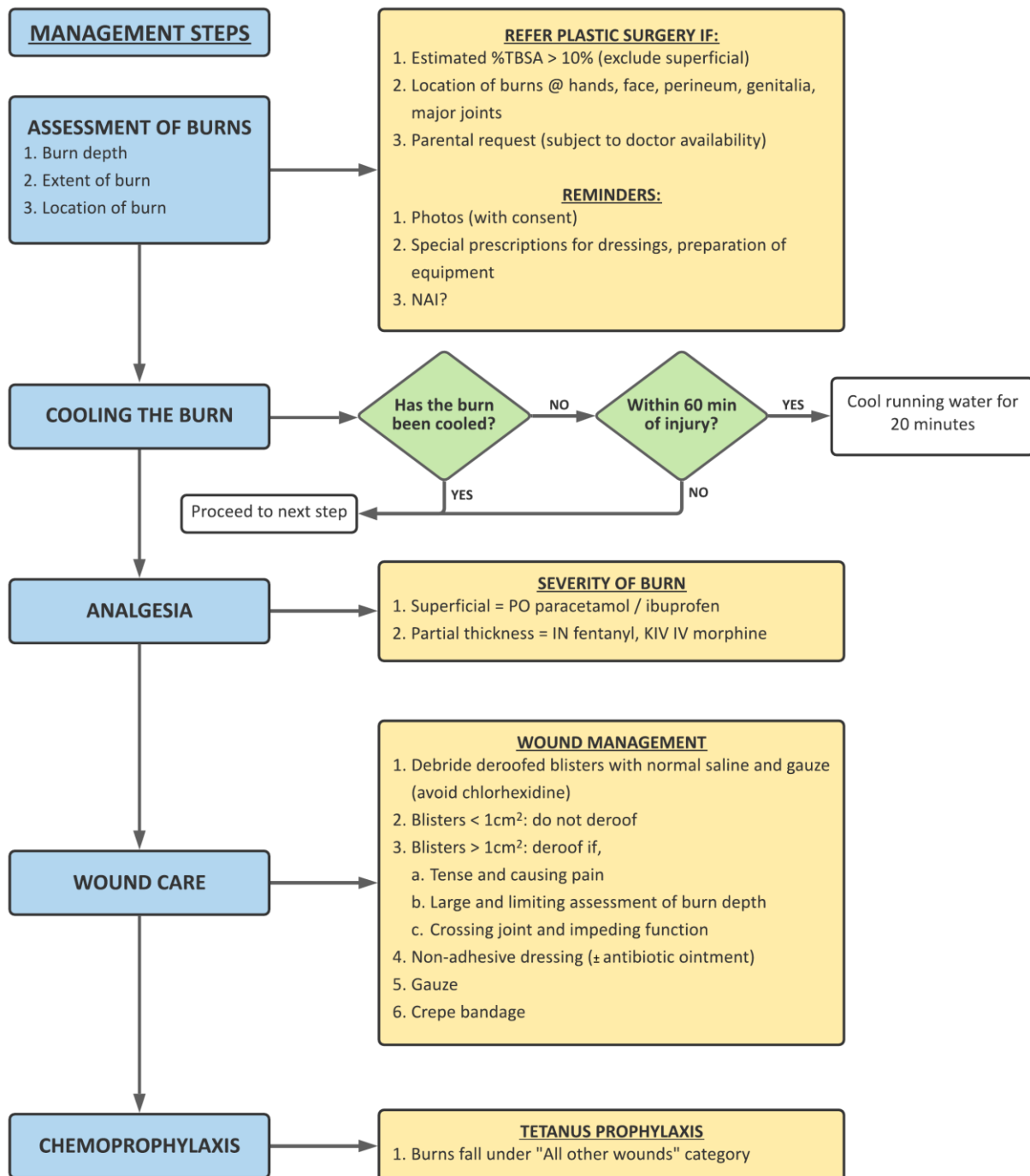
Area	0 – 1 year	1 year	5 year	10 year	15 year	Adult
A = ½ of head	9.5%	8.5%	6.5%	5.5%	4.5%	3.5%
B = ½ of thigh	2.75%	3.25%	4%	4.5%	4.5%	4.75%
C = ½ of lower leg	2.5%	2.5%	2.75%	3%	3.25%	3.5%

PALM METHOD

- Palm + Fingers = 1%
- Palm only = 0.5%



MANAGEMENT OF MINOR – MODERATE BURNS IN THE CHILDREN'S EMERGENCY



FIRST AID

1. Stop the burn progression
 - Remove clothing and jewelry
2. Cool the burn
 - Run under cool water for at least 20 minutes within 60 minutes of injury
 - Do not apply toothpaste, ointments, ice or iced water
3. Keep patient warm

BURNS MANAGEMENT IN THE CHILDREN'S EMERGENCY

ASSESS SEVERITY OF BURN

- Depth of burn, %TBSA, location of burn
- Indications for plastic surgery referral:
 - i. Partial-thickness burn > 10% TBSA
 - ii. Full thickness burns
 - iii. Special areas involved (hands, feet, face, and perineum)
 - iv. Significant electrical or chemical burns
- Could this be a non-accidental injury (NAI)?
 - i. Delayed presentation
 - ii. Inconsistent history and physical examination findings
 - iii. Suspicious burn pattern (e.g., Lack of splash marks in scalding injury, pattern-burns etc.)
 - Bilateral lower limb burns
 - Pattern-burns (well defined burn patterns)
 - Lack of splash marks in scalding injury
 - Bilateral lower limb burns

COOL THE BURN

- If first aid has been performed before arrival, skip this step in CE
- Cool the burn with cool running water for at least 20 minutes (if not already done) if patient presents within 60 minutes from burn injury
- This will help with pain experienced by patient as well
- This is balanced against development of hypothermia (especially in younger patients)

ANALGESIA

- Superficial burns = PO paracetamol or PO ibuprofen
- Partial thickness (or deeper) = IN fentanyl (KIV IV morphine)

WOUND CARE

- Cleansing with normal saline and gauze to remove sloughing epidermis; avoid chlorhexidine
- Intact blisters < 1cm² should not be deroofed
- For intact blisters > 1cm², consider deroofing blister if:
 - i. Blister is tense and exerting pressure on the nerve endings in dermis, resulting in pain
 - ii. Blister crosses a joint, impeding range of movement and function
 - iii. Blister affecting assessment of depth of burn
- Deroofing blisters could:
 - i. Exacerbate pain by exposing nerve endings to environment
 - ii. Create a portal of entry for infection
- Dressing layers:
 - i. Non-occlusive dressing (± 1% chlortetracycline ointment "Latycin")
Only **UrgoTül** is available in CE (prescription needed for other non-adhesive dressings requested by plastic surgery doctor)
 - ii. Gauze
 - iii. Crepe bandage

CHEMOPROPHYLAXIS

- Minor wounds involving violation of the epidermis is not considered clean

Table 1 - "Red Book®: 2015 Report of the Committee on Infectious Diseases, 30th Edition"

		Clean / Superficial Wounds		All other wounds ¹	
		DTaP, Tdap, or Td	TIG ²	DTaP, Tdap, or Td	TIG ²
History of adsorbed tetanus toxoid vaccine	< 3 doses or unknown	Yes	No	Yes	Yes
	≥ 3 doses	No (< 10y since last dose)	No	No (< 5y since last dose)	No
		Yes (≥ 10y since last dose)	No	Yes (≥ 5y since last dose)	No

1: All other wounds = Wounds contaminated with dirt, feces, soil & saliva; puncture wounds; fracture avulsions; wounds from missiles, crushing, burns, and frostbite.

2: TIG = Tetanus Immunoglobulin (Human); IVIg can be used if TIG is not available

DISPOSITION

- Discharge criteria:
 - Uncomplicated minor burns
 - Patients referred to plastic surgery cleared by surgeons for discharge
 - No systemic complications or associated blunt trauma
 - No suspicion of NAI
 - Parents confident of care
- Discharge advice:
 - Keep dressing clean and dry
 - Do not remove dressing or attempt to change dressing at home
 - Risk of burns conversion over the 1st week
 - Return to CE and/or PRAS clinic if symptoms of wound infection (fever, wound discharge)
- Follow up:
 - TCU PRAS clinic (Monday, Wednesday, Friday) for first dressing change and wound inspection, 2 – 3 days after discharge
 - Subsequent regular dressing changes 2x per week @ Surgical Specialist Clinic, Level 15, Corridor B, Medical Centre

PATHOPHYSIOLOGY OF BURNS

LOCAL RESPONSE

- Direct coagulation of skin and subcutaneous fat
- Microvascular vasoconstriction and thrombosis → burns appear to deepen in the days after initial evaluation "burn progression"

SYSTEMIC RESPONSE

- Release of cytokines and inflammatory mediators
- Early hypodynamic state (first 48H)
 - Capillary leak
 - Neurohormonal mediated change in vessel tone
 - Depressed myocardial function (TNF-α)
- Hyperdynamic state (> 48H)
 - Increased cardiac output and raised baseline metabolic rate
 - Enhanced gluconeogenesis, protein catabolism and insulin resistance

MANAGEMENT OF MODERATE – SEVERE BURNS IN THE CHILDREN’S EMERGENCY

ASSESS SEVERITY AND MECHANISM OF BURNS

- Depth and extent of burn
 1. Burns > 10% TBSA require PRAS referral and oral hydration
 2. Burns > 20% TBSA is at risk of hypotension from early systemic inflammatory response with capillary leak; will require IV hydration and careful titration of fluid requirements
- Type of burn → refer to <Special Considerations>

PRIMARY SURVEY

AIRWAY / BREATHING

- Evidence of inhalational injury: singed nasal hair, productive cough, carbonaceous sputum, stridulous breathing, increased work of breathing
- Possible impending airway compromise – airway edema peaks 12 – 36H post-injury
- Consider an x-ray radiograph of the chest and a blood gas if SpO₂ < 95% on RA, if oxygen requirements > 2L/min, if there are lower tract signs

CIRCULATION

- Patients who are hypotensive and in shock should be resuscitated with IV crystalloids in aliquots of 10 – 20ml/kg
- “Fluid creep” – overzealous fluid resuscitation resulting in significant 3rd spacing with possible compromise of an already edematous airway or worsening of compartment syndrome

FLUID RESUSCITATION / PARKLAND FORMULA

- Patients with burns > 20% TBSA should be resuscitated with IV fluids; Hypotension develops secondary to capillary leak driven by systemic inflammatory response
- Parkland Formula: $Volume (ml) = 4ml/kg/\%TBSA$
 - Administer half this volume over first 8 hours, and second half over the next 16 hours
 - Timing should begin from time of injury instead of time of arrival to CE
- Fluid choice = Lactated Ringers or Normal Saline
- Add maintenance fluids containing D5% for younger patients

THERAPEUTIC ENDPOINTS:

- Urine output 1 – 2ml/kg/h
- Reversal of abnormal vital signs

DISABILITY + EXPOSURE

- Differentials for patients with altered mental status include:
 - Intracranial hemorrhage secondary to associated blunt trauma
 - Electrical injury with CNS complications
 - Carbon monoxide or hydrogen cyanide poisoning; suspected if risk factor of flame injury and initial assessment (history, physical examination and POCT) cannot explain degree of AMS with ± multi-systemic dysfunction
- Prevent hypothermia

WOUND CARE

1. Analgesia
2. Cleansing wound + debridement + blister management
3. Dressing wound
4. Chemoprophylaxis

SPECIAL CONSIDERATIONS (FOR EACH BURN TYPE)

THERMAL

1. Scald
2. Contact
 - Significant burns result from prolonged contact (spinal reflex to withdraw normally)
 - Evaluate for reason: altered mental state (seizures, drugs, LOC etc), NAI
3. Flame
 - House fire
 - Associated with inhalational injuries & blunt trauma
 - Increased risk of carbon monoxide and hydrogen cyanide poisoning (suspect when child has altered mental status with no apparent cause with multi systemic involvement)
4. Electrical (see below)
5. Others:
 - Sun burn
 - Friction:
 - Common mechanisms: Treadmill injury & Vacuum-cleaner injury
 - Treadmill: Older patient (3-years), dominant hand, volar >> dorsal
 - Vacuum: Younger patients (18-months), no hand dominance, dorsal > volar

CARBON MONOXIDE POISONING

- Pathophysiology
 - Competes with O₂ for binding to Hb → reduced O₂ carrying capacity → functional anemia
 - Alters structure and prevents off-loading of O₂ at other binding sites → left shift of oxy-hemoglobin curve
 - Results in tissue hypoxia & conversion to anaerobic metabolism
- Symptoms are non-specific:
 - Constitutional – Nausea, malaise, dizziness
 - CNS – Seizures, syncope, coma
 - CVS/Respi – Myocardial ischemia, arrhythmia, pulmonary edema
 - Metabolic – Lactic acidosis
- Consider carbon monoxide poisoning in an obtunded child with evidence of multi-organ hypoxia with no obvious cause → normal pulse oximetry and ABG (pO₂ is unchanged – plasma O₂ content) and does not respond to supplemental O₂
- Check blood carboxy-hemoglobin levels (normal CO-Hb 3%, smokers CO-Hb 10 – 15%)
- Treatment = FiO₂ 100%

HYDROGEN CYANIDE POISONING

- Pathophysiology: inhibits aerobic metabolism → anaerobic metabolism → lactic acidosis
- Sources of cyanide:
 - Household fires (wool, silk, nylon, plastics, melamine resins)
 - Industrial exposure (plastic production, synthetic rubber, metal polish, fertilizer, rodenticide)
 - Miscellaneous (cigarette smoking, artificial glue sniffing, macadamia nuts, bamboo sprouts)
- Symptoms are non-specific (most common = CNS + CVS)
 - CNS – Headache, anxiety, confusion, LOC, seizures
 - CVS – Tachycardia and hypertension → bradycardia and hypotension → dysrhythmias
 - GI – Abdominal pain, vomiting, hepatic necrosis
 - Renal – Renal failure
 - Lung – Tachypnoea, bradypnea, pulmonary oedema
- Consider cyanide poisoning if severity of patient's symptoms not explained by CO-Hb levels

- Investigations: lactate, venous blood gas for venous hyperoxia (pO₂) – inability for end organs to use oxygen → higher venous pO₂

ELECTRICAL

- Electric sockets in Singapore provide low voltage (230V) with alternating currents (AC); with Residual Current Circuit Breaker (RCCB) tripping when current drawn crosses capacity of circuit breaker
- Mechanism
 - Direct effect of electric injury on tissue
 - Conversion of electrical energy to thermal energy
 - Associated blunt trauma (e.g., fall, muscle contracture)

	ARC INJURY	LOW VOLTAGE	HIGH VOLTAGE	LIGHTNING
Voltage	< 1000 V	< 1000 V	>1000 V	> 30 x 10 ⁶ V
Type of current	AC	AC	AC / DC	DC
Duration	Brief	Prolonged	Brief	Instantaneous
Burns	Common	Uncommon, superficial	Common, deep	Uncommon
Mortality	Low	Low	Moderate	High
Example	Home appliances	Home socket	Electric cables	

COMPLICATIONS

- CNS Mono/polyneuropathies
Keraunoparalysis – pulseless, blue, mottled limb (LL > UL)
- CVS Arrhythmias – ventricular fibrillation (AC), asystole (DC)
- Respi Tetany of respiratory muscles
Central control of breathing
- MSK Muscle edema, rhabdomyolysis, compartment syndrome, myonecrosis, osteonecrosis
- Renal Pigment induced nephropathy
- Haem DIVC
- Ears/Eyes TM rupture, globe injuries from thunder

LIGHTNING

	MECHANISM
Direct strike	Lightning strikes patient
Contact injury	Lightning strikes object patient in contact with
Side flash	Current splashes from object
Ground current	Lightning strikes ground → conducted through ground into patient
Upward streamer	Lightning does not strike the ground; current conducted from ground through patient to meet negative charge of stepped leader
Blast injury	Blast effect (thunder) resulting in ruptured TM or blunt trauma



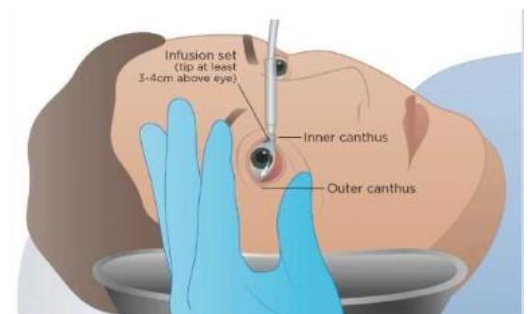
- Pathognomonic skin changes of lightning strike – Lichtenberg Figures (Fern-like burn pattern, appears after 1H and disappears by 24 – 36H)

CHEMICAL

- Caustic and corrosive substances found in most homes (detergents, toilet cleaners)
- Acidic substances – result in self-limiting burns, forming coagulum which limits spread
- Alkaline substances – results in extensive, penetrative burns as it dissolves tissue, resulting in fluid loss; more common and often deleterious
- General approach to chemical burns:
 1. PPE for HCW
 2. Remove patient from area of exposure
 3. Remove jewelry and clothing
 4. Brush off dry chemicals
 5. Irrigate with copious amounts of water or isotonic solution until pH is neutral (pH 6.5 – 7.5)
- Exceptions:
 1. Dry lime: adding water creates a strong alkali
 2. Phenol: more readily absorbed through skin when water added, wipe with sponges soaked in polyethylene glycol (PEG)
 3. Elemental metals: combustion with addition of water, mineral oil buffer
 4. Hydrofluoric Acid
 - Binding properties of fluoride can cause systemic hypocalcemia and hypomagnesemia even with 1% TBSA burns
 - Topical application of calcium gluconate 2.5% gel deactivates fluoride ion

CHEMICAL INJURY TO THE EYE

- Obtain pH with litmus paper before topical analgesia with tetracaine (acidic)
- Provide analgesia with topical tetracaine 0.5% (effect lasts 30 minutes)
- Start irrigation until pH = 7
- Refer Eye on-call



HOW TO IRRIGATE EYE?

- Position patient and support head, placing kidney dish against ipsilateral cheek to catch run off fluid
- Position infusion tubing 3 – 4 cm from eye, directing flow with infusion tubing
- Irrigate eye from inner canthus to outer canthus
- Check eye pH every 15 minutes until pH = 7

REFERENCES

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